

Case Study

Getsafe Insurance · Data Team

Part I · Data Modeling & Accounting Reconciliation

Part II · Customer KPI Data Mart Design

Refer: <https://tinyurl.com/5b2f7jj8>

2,660

Raw Transactions

4

Insurance Partners

3

Months Analyzed

€ 17,443.22

Total Accounting Premium

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Agenda

01 Business Context

Why this case study matters

02 Medallion Architecture

Bronze → Silver → Gold with DuckDB & dbt

03 Data Quality Findings

What we found in the raw data

04 Part I: Monthly Premiums

Building the Finance data mart

05 Reconciliation Analysis

Finance vs Accounting — full picture

06 Part II: KPI Data Mart

Daily customer activity for BI tools

07 dbt Implementation

Production transformation layer

08 Recommendations

Roadmap and next steps

Business Context

The Problem

- Accounting closes monthly premiums manually from invoices
- Finance team wants the same numbers from transaction records
- No automated comparison → manual spreadsheet reconciliation
- No scalable BI layer for customer KPIs (active count, premium)

The Approach

- Medallion Architecture: Bronze → Silver → Gold
- Automated reconciliation with root cause analysis
- Date-spine KPI mart — all 3 KPIs via SUM/COUNT only
- Production dbt models with tests and documentation

SO WHAT?

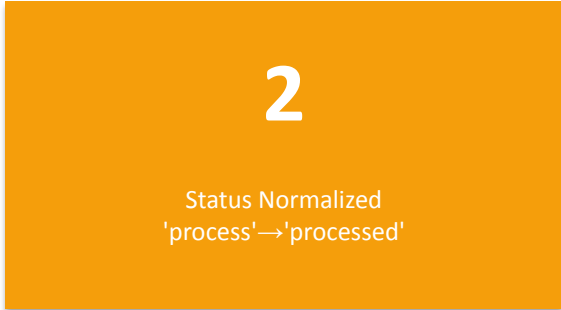
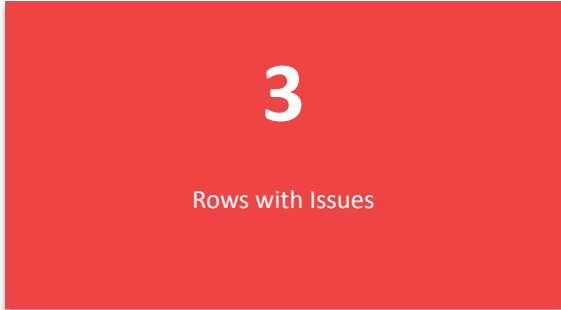
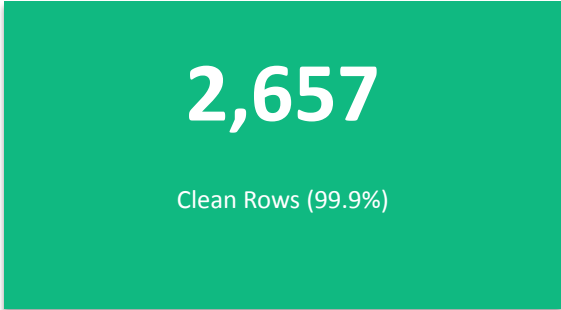
A reusable data pipeline eliminates manual monthly reconciliation, creates a single source of truth for Finance and Accounting, and enables self-serve BI analytics.

Medallion Architecture

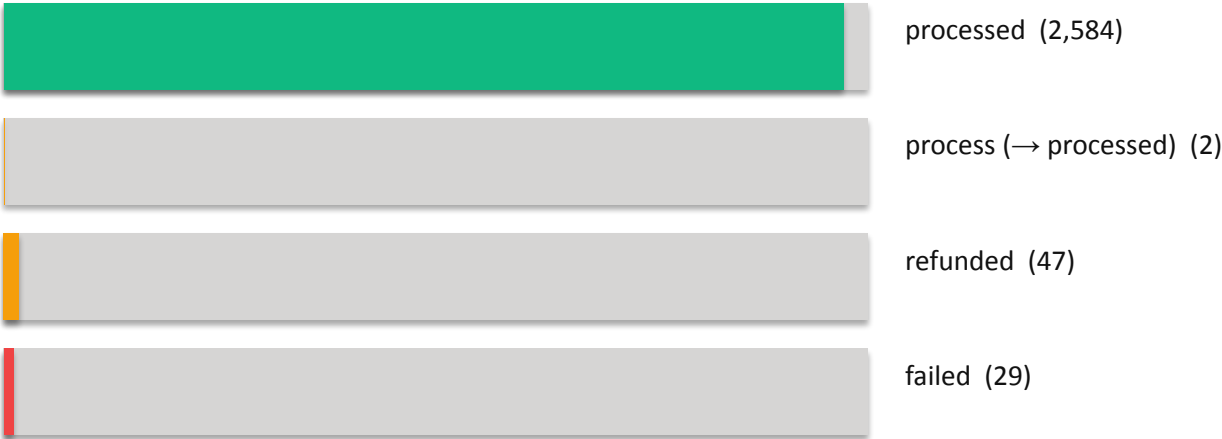


dbt-duckdb translates all transformation logic into tested, versioned SQL models

Data Quality Findings — Silver Layer



Status Distribution



Issues Found

- negative_premium — 3 rows**
Excluded from Gold via `_is_clean = FALSE`
- 'process' status — 2 rows**
Normalized to 'processed' in Silver — included in premium sums
- Sep 2025 rows — 3 rows**
Outside accounting period — in the data but outside reconciliation window

SO WHAT? Data quality is 99.9% — the pipeline is production-ready. The 3 flagged rows are preserved (not dropped) for full auditability.

Part I — Monthly Premium Data Mart

Party	Jun 2025	Jul 2025	Aug 2025	Total
Berlinre	€ 1,476.45	€ 1,084.40	€ 714.06	€ 3,274.91
Dronant	€ 2,983.41	€ 1,771.90	€ 1,100.99	€ 5,856.30
Getland	€ 1,730.31	€ 1,412.28	€ 660.19	€ 3,802.78
Liadigital	€ 2,239.33	€ 1,353.47	€ 878.49	€ 4,471.29
TOTAL	€ 8,429.50	€ 5,622.05	€ 3,353.73	€ 17,405.28

Filter: status = 'processed' AND premium_amount > 0 AND _is_clean = TRUE

Reconciliation: Finance vs Accounting

7/12

Exact Matches

1/12

Discrepancies

€ 37.94

Total Delta

0.28%

Delta / Total Premium

Party	Month	Accounting	Finance	Delta	Status
berlinre	2025-06	€ 1,483.22	€ 1,476.45	€ -6.77	NEAR_MATCH
berlinre	2025-07	€ 1,098.74	€ 1,084.40	€ -14.34	DISCREPANCY
berlinre	2025-08	€ 714.06	€ 714.06	€ +0.00	MATCH
dronant	2025-06	€ 2,983.41	€ 2,983.41	€ +0.00	MATCH
dronant	2025-07	€ 1,780.19	€ 1,771.90	€ -8.29	NEAR_MATCH
dronant	2025-08	€ 1,107.07	€ 1,100.99	€ -6.08	NEAR_MATCH
getland	2025-06	€ 1,730.31	€ 1,730.31	€ +0.00	MATCH
getland	2025-07	€ 1,412.28	€ 1,412.28	€ +0.00	MATCH
getland	2025-08	€ 660.19	€ 660.19	€ +0.00	MATCH
liadigital	2025-06	€ 2,241.79	€ 2,239.33	€ -2.46	NEAR_MATCH
liadigital	2025-07	€ 1,353.47	€ 1,353.47	€ +0.00	MATCH
liadigital	2025-08	€ 878.49	€ 878.49	€ +0.00	MATCH

Discrepancy Root Cause Analysis

1 Timing Cutoff

Accounting closes on invoice date; Finance uses transaction timestamp. Transactions near month-end may fall in different periods.

2 Missing Processed Rows

Finance shows lower sums for berlinre & dronant. Possible data extract issue: some transactions not captured in the CSV export window.

3 Refund Netting

47 refund rows totalling EUR 328 are excluded from Finance sums but may be treated differently in Accounting.

4 Data Extraction Window

The CSV covers Jun 1 – Aug 4, 2025. If accounting month-end is Aug 31, late-August transactions are missing from the Finance source.

Best filter: status = 'processed' only · Total delta: EUR 37.94 · 0.28% of total accounting premium — within 1% materiality threshold

Part II — KPI Data Mart: Design Approach

The constraint: BI tool can only use SUM and COUNT — but checking if a customer is active on date X requires knowing if X falls between started_at and churned_at. That's a range join. You can't do a range join with SUM or COUNT alone.

Solution — Date Spine: Pre-explode each customer's active window into one row per day. Then every KPI is just SUM(daily_premium) or COUNT(user_id) with a WHERE date filter.

Input → Output

1 row: USR001 | started=Jan 1 | churned=Jan 3 | premium=€30/mo

→ Jan 1 | USR001 | daily_premium=0.968
→ Jan 2 | USR001 | daily_premium=0.968
→ Jan 3 | USR001 | daily_premium=0.968

KPI 1	KPI 2	KPI 3
Active customers/day	Premium in date range	Accumulated premium
COUNT(DISTINCT user_id) WHERE date = X GROUP BY product_group	SUM(daily_premium) WHERE date BETWEEN A AND B	SUM(daily_premium) WHERE date <= target

Retroactive Churn

- Churned_at moves to the past → customer was never active after that date
- Fix: DELETE rows WHERE user_id=X AND activity_date > new_churned_at
- dbt: delete+insert incremental — only the affected customer is reprocessed
- Unique key: (user_id, activity_date) makes the operation idempotent

Scalability

- 1M customers × 3yr avg tenure → ~1.1B rows
- Partition by activity_month — BI date filters skip irrelevant partitions
- Incremental run: only process customers with updated_at > last run
- Optional: summary layer (date, product_group) for dashboard performance

dbt Implementation

Model	Layer	Materialization	Description
bronze_raw_transactions	BRONZE	TABLE	Provenance + schema tests
silver_transactions	SILVER	TABLE	Cleaned, normalized, DQ-flagged
snp_silver_transactions	SNAPSHOT	SNAPSHOT	SCD Type 2 — point-in-time audit
gold_fct_monthly_premiums	GOLD	TABLE	Written + earned premium per party/month
gold_fct_accounting_reconciliation	GOLD	TABLE	Finance vs Accounting report
gold_fct_customer_activity_daily	GOLD	INCREMENTAL	Date-spine KPI mart

Tests: not_null · unique · accepted_values · dbt_utils.unique_combination_of_columns · Seeds: accounting_closing.csv · Profiles: dev + prod

Recommendations & Roadmap

SHORT-TERM (< 2 weeks)

Align data extraction window: Finance extract must run after month-end close

Agree on official status filter definition — document in dbt as a variable

Add uniqueness constraint at the billing API to prevent duplicate transaction IDs

MEDIUM-TERM (< 2 months)

Add `net_premium` column (gross – refunds) to `fct_monthly_premiums`

Connect Gold marts to BI tool; alert on delta > EUR 0.01 post month-close

Refine earned premium: join policy dimension for actual `policy_start_date`

LONG-TERM (< 6 months)

Build `gold_fct_loss_ratio` — blocked on `claims_payments` from Claims team

Eliminate manual invoice process — Gold model as the single source of truth

Connect Part II to real customer data — model is production-ready

Business value: eliminates manual reconciliation (~8h/month), reduces audit risk, and enables data-driven decisions at scale.

THANK YOU

Questions & Discussion

- Pipeline:** Bronze → Silver → Gold · Python + DuckDB + dbt
- Part I:** EUR 37.94 delta (0.28%) · Root causes documented · 4 recommendations
- Part II:** Date-spine mart · 3 KPIs via SUM/COUNT · Retroactive-change-safe
- Code:** Idempotent · dbt tested · All models documented